BM and GAUSSIAN PROCESSES

— Recall on Gaussian r.v.'s —

A 1-dim. r.v. Γ is Gaussian if it has the characteristic function

$$\mathbb{E}\, e^{i\xi\Gamma} = e^{im\xi - \frac{1}{2}\sigma^2\xi^2} \qquad 1)$$

for some real numbers $m \in \mathbb{R}$ and $\sigma > 0$.

By differentiating 1) two times it is easy to see that

$$m = \mathbb{E}\,\Gamma \qquad \sigma^2 = \mathrm{Var}\,\Gamma$$

A random vector $(\Gamma_1 \dots \Gamma_n) =: \underline{\Gamma} \in \mathbb{R}^n$ is Gaussian if $\langle \underline{\ell}, \underline{\Gamma} \rangle$ is, for every $\underline{\ell} \in \mathbb{R}^n$ a 1-dim Gaussian.

This is the same requiring that

$$\mathbb{E}\, e^{i\langle \underline{\xi}, \underline{\Gamma}\rangle} = e^{i\,\mathbb{E}\langle \underline{\xi}, \underline{\Gamma}\rangle - \frac{1}{2}\mathrm{Var}\langle \underline{\xi}, \underline{\Gamma}\rangle}. \qquad 2)$$

Setting $\underline{m} = (m_1 \dots m_n) \in \mathbb{R}^n$ and $\Sigma := (\Sigma)_{jk} = \sigma_{jk}$, $j, k = 1 \dots n$ where

$$m_j := m_j \quad \text{and} \quad \sigma_{jk} := \mathbb{E}(\Gamma_j - m_j)(\Gamma_k - m_k) = \mathrm{Cov}(\Gamma_j \Gamma_k)$$

$\mathrm{Var}\langle \underline{\xi}, \underline{\Gamma}\rangle = \mathrm{Var}\, \xi_1 \Gamma_1 + \ldots + \mathrm{Var}\, \xi_n \Gamma_n$ + all covariances

it possible to rewrite 2) as

$$\mathbb{E}\, e^{i\langle \underline{\xi}, \underline{\Gamma}\rangle} = e^{i\langle \underline{\xi}, \underline{m}\rangle - \frac{1}{2}\langle \underline{\xi}, \Sigma \underline{\xi}\rangle}.$$

The vector $\underline{m}$ is called the mean vector, the matrix Σ is called the variance and covariance matrix.

We first obtain some properties of BM following from the definition which are related with the properties of the Gaussian r.v.'s.

—— 1-dim BM

It follows by definition that $\mathbb{E}\, e^{i\xi B_t} = e^{-\frac{t}{2}\xi^2}$ for all $t \geq 0$, $\xi \in \mathbb{R}$. It is possible to see that the integral $\mathbb{E}\, e^{i\xi B_t}$ is convergent for any $\xi \in \mathbb{C}$. Indeed, take $\mathrm{Re}\,\xi = 0$, then

$$\int_{-\infty}^{+\infty} e^{i\xi x} \frac{e^{-\frac{x^2}{2t}}}{\sqrt{2\pi t}}\, dx = \int_{-\infty}^{+\infty} e^{\overbrace{-\mathrm{Im}(\xi)}^{=:c} x} \frac{e^{-\frac{x^2}{2t}}}{\sqrt{2\pi t}}$$

$$= \int_{-\infty}^{+\infty} e^{cx} \frac{e^{-\frac{x^2}{2t}}}{\sqrt{2\pi t}}\, dx \leq \int_{-\infty}^{+\infty} e^{c^2} \frac{e^{-\frac{x^2}{4t}}}{\sqrt{2\pi t}}\, dx < +\infty$$

where we used that

$$e^{(y/2-c)^2} \geq 1 \longrightarrow e^{cy} \leq e^{c^2+y^2/4} \longrightarrow e^{cy} e^{-y^2/2} \leq e^{c^2} e^{-y^2/4}.$$

It follows that

$$\mathbb{E}\, e^{zB_t} = e^{t z^2/2} \quad \text{for all } z \in \mathbb{C}.$$

Moments of BM can be obtained with a direct computing. Indeed, for $k = 0, 1, 2, \ldots$

$$\mathbb{E}\, B_t^{2k+1} = \int_{-\infty}^{+\infty} \underbrace{x^{2k+1} \frac{e^{-\frac{x^2}{2t}}}{\sqrt{2\pi t}}}_{\text{odd}}\, dx = 0$$

$$\mathbb{E}\, B_t^{2k} = \int_{-\infty}^{+\infty} x^{2k} \frac{e^{-\frac{x^2}{2t}}}{\sqrt{2\pi t}}\, dx$$

$$= 2 \int_0^{\infty} x^{2k} \frac{e^{-\frac{x^2}{2t}}}{\sqrt{2\pi t}}\, dx \qquad \begin{cases} \frac{x^2}{2t} = y \\ x = \sqrt{2ty} \\ dx = \sqrt{2t}\, \frac{1}{2}\, y^{-1/2} \end{cases}$$

$$= \cancel{2} \int_0^{\infty} (2ty)^k e^{-y} \frac{1}{\sqrt{\cancel{2}\pi \cancel{t}}}\, \cancel{\sqrt{2t}}\, \frac{1}{\cancel{2}}\, y^{-1/2}\, dy$$

$$= 2^k t^k \frac{1}{\sqrt{\pi}} \int_0^{\infty} y^{k-1/2} e^{-y}\, dy$$

$$= \frac{2^k t^k}{\sqrt{\pi}} \int_0^{\infty} y^{k+1/2-1} e^{-y}\, dy$$

$$= \frac{2^k t^k}{\sqrt{\pi}} \Gamma(k+1/2).$$

It follows in particular that

$$E B_t = E B_t^3 = E B_t^5 \cdots = 0$$
$$\operatorname{Var} B_t = E B_t^2 = t \qquad E B_t^4 = 3t^2.$$

$$\operatorname{Cov}(B_s, B_t) = E B_s B_t = s \wedge t$$

↳ to see this, let $s \le t$,

$$E B_s B_t = E\left(B_s(B_t - B_s) + B_s^2\right)$$

indep of increm.
$$= \underbrace{E B_s}_{0} \underbrace{E(B_t - B_s)}_{0} + \underbrace{E B_s^2}_{s} = s.$$

Definition. A 1-dim stochastic process $(X_t)_{t\ge 0}$ is called a Gaussian process if all vectors

$$(X_{t_1}, \cdots X_{t_n}),\ n \ge 1,\ 0 \le t_1 \le t_2 \le \cdots \le t_n$$

are Gaussian (possibly degenerate).

Theorem. A 1-dim BM is a Gaussian process. Further for $t_0 := 0 < t_1 < \cdots < t_n$, $n \geq 1$, the vector

$$(B_{t_1} \cdots B_{t_n})^T =: \underline{\Gamma}$$

is a Gaussian r.v. with variance covariance matrix C: $(C)_{jk} = (t_j \wedge t_k)_{jk=1\dots n}$ and $\underline{m} = 0$. Moreover it admits the density function

$$\frac{1}{(2\pi)^{n/2}} \frac{1}{\sqrt{\det C}} e^{-\frac{1}{2}\langle x, C^{-1} x\rangle} = \frac{1}{(2\pi)^{n/2}\sqrt{\prod_{j=1}^{n}(t_j - t_{j-1})}} e^{-\frac{1}{2}\sum_{j=1}^{n} \frac{(x_j - x_{j-1})}{t_j - t_{j-1}}}$$

Proof. We should prove that

$$\mathbb{E}\, e^{i\langle \underline{\xi}, \underline{\Gamma}\rangle} = e^{-\frac{1}{2}\langle \underline{\xi}, C\underline{\xi}\rangle}$$

Observe that

$$\underline{\Gamma} = \overbrace{\begin{pmatrix} 1 & 0 & \cdots & 0 \\ 1 & 1 & & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & & 0 \\ 1 & 1 & \cdots & 1 \end{pmatrix}}^{=:M}_{n\times n \text{ lower triangular}} \overbrace{\begin{pmatrix} B_{t_1} - B_{t_0} \\ \vdots \\ B_{t_n} - B_{t_{n-1}} \end{pmatrix}}^{=:\Delta}_{n\times 1}$$

since

$$B_{t_k} - B_{t_0} = \sum_{j=1}^{k} (B_{t_j} - B_{t_{j-1}}) \qquad \begin{pmatrix}\text{recall} \\ t_0 = 0\end{pmatrix}$$

Hence

$$E e^{i<\xi, \Gamma>} = E e^{i<\xi, M\Delta>}$$

$$= E e^{i<M^T\xi, \Delta>} \qquad M^T \overset{\xi}{\cancel{\Delta}} = \begin{pmatrix} \xi_1+\dots+\xi_m \\ \xi_2+\dots\xi_m \\ \vdots \\ \xi_m \end{pmatrix}$$

$$= E e^{i\sum_{j=1}^{m} (B_{t_j}-B_{t_{j-1}})(\xi_j+\dots+\xi_m)}$$

indep. of increm. $$= \prod_{j=1}^{m} E e^{i(B_{t_j}-B_{t_{j-1}})(\xi_j+\dots+\xi_m)}$$

Gaussianity $$= \prod_{j=1}^{m} e^{-\frac{1}{2}(t_j-t_{j-1})(\xi_j+\dots+\xi_m)^2}$$

$$= e^{-\frac{1}{2}\left[\sum_{j=1}^{m} t_j(\xi_j+\dots+\xi_m)^2 - \sum_{j=1}^{m} t_{j-1}(\xi_j+\dots+\xi_m)^2\right]}$$

Note that

$$\sum_{j=1}^{m} t_j(\xi_j+\dots+\xi_m)^2 - \sum_{j=1}^{m} t_{j-1}(\xi_j+\dots+\xi_m)^2$$

$j'=j-1 \qquad j=j'+1$

$$= \overbrace{t_m\xi_m^2}^{\text{last term}} + \sum_{j=1}^{m-1} t_j(\xi_j+\dots+\xi_m)^2 - \sum_{j'=0}^{m-1} t_{j'}(\xi_{j+1}+\dots+\xi_m)^2$$

↳ the term $j'=0$ is zero since $t_{j'}=0$ $\quad (j'=0)$

$$= t_m\xi_m^2 + \sum_{j=1}^{m-1} t_j\left[\underbrace{(\xi_j+\dots+\xi_m)^2 - (\xi_{j+1}+\dots+\xi_m)^2}\right]$$

$$\xi_j^2+\dots+\xi_m^2+\sum_{\substack{k\neq h \\ j\le k,h\le m}}\xi_k\xi_h - \xi_{j+1}^2 \dots - \xi_m^2 - \sum_{\substack{k\neq h \\ j+1\le k,h\le m}}\xi_k\xi_h$$

$$= \xi_j^2 + 2\sum_{k=j+1}^{m}\xi_j\xi_k = \xi_j\left(\xi_j+2\sum_{k=j+1}^{m}\xi_k\right)$$

$$= t_m\xi_m^2 + \sum_{j=1}^{m-1} t_j\,\xi_j\,(\xi_j+2\xi_{j+1}+\dots+2\xi_m)$$

$$= t_m\xi_m^2 + t_1\xi_1^2 + 2t_1\xi_1\xi_2 + \dots + 2t_1\xi_1\xi_m$$

$$+ \cdots + t_{n-1} \xi_{n-1}^2 + t_{n-1} \xi_{n-1} \xi_n$$

$$= \sum_{j=1}^{n} \sum_{k=1}^{n} (t_j \wedge t_k) \xi_j \xi_k$$

Hence $\quad \mathbb{E}\, e^{i \langle \underline{\xi}, \underline{\Gamma} \rangle} = e^{-1/2 \langle \underline{\xi}, C \underline{\xi} \rangle}$.

where

$$C = \begin{bmatrix} t_1 & t_1 & \cdots & t_1 \\ t_1 & t_2 & t_2 \cdots & t_2 \\ \vdots & t_2 & \ddots & \\ \vdots & \vdots & & \\ t_1 & t_2 & & t_n \end{bmatrix}$$

↳ strictly positive definite and symmetric

↳ its inverse C^{-1} ∃ and is again positive definite.

In order to see that Γ admits the density above, it is possible to directly compute

$$\int_{\mathbb{R}} e^{i \langle x, \xi \rangle} (2\pi)^{-\frac{n}{2}} (\det C)^{-1/2} e^{-1/2 \langle x, C^{-1} x \rangle} \, dx$$

$$\cdots = e^{-\frac{1}{2} \langle \xi, C \xi \rangle}$$

and it can be directly computed that

$$\langle x, C^{-1} x \rangle = \sum_{j=1}^{n} \frac{(x_j - x_{j-1})^2}{t_j - t_{j-1}}$$

$\lrcorner$

Inspection of the proof above actually shows more than the fact the BM is a Gaussian process.

Theorem. Let $(X_t)_{t\geq 0}$ be a 1-dim Gaussian process s.t. $\underline{\Gamma} := (X_{t_1}, \dots, X_{t_n})^T$ is Gaussian with mean $\underline{0}$ and covariance $C : (C)_{jk} = (t_j \wedge t_k)_{j,k=1,\dots,n}$. If $(X_t)_{t\geq 0}$ has continuous sample paths, then $(X_t)_{t\geq 0}$ is a 1-dim BM.

Proof. Continuity of paths is assumed. Then $B_0 \sim N(0,0) = \delta_0$. Set $\Delta = (X_{t_1} - X_{t_0}, \dots, X_{t_n} - X_{t_{n-1}})$ and let M be as above, so that, as above

$$\underline{\Gamma} = M\Delta \quad \Rightarrow \quad \Delta = M^{-1}\underline{\Gamma}$$

We directly compute

$$\mathbb{E}\, e^{i\langle \xi, \Delta\rangle} = \mathbb{E}\, e^{i\langle \xi, M^{-1}\underline{\Gamma}\rangle} = \mathbb{E}\, e^{i\langle (M^{-1})^T\xi, \underline{\Gamma}\rangle}$$

$\underline{\Gamma}$ Gaussian
$$= e^{-\frac{1}{2}\langle (M^{-1})^T\xi, C(M^{-1})^T\xi\rangle}$$
$$= e^{-1/2\langle \xi, M^{-1}C(M^{-1})^T\xi\rangle}$$

One can see that

$$M^{-1}\Sigma(M^{-1})^T = \begin{pmatrix} t_1-t_0 & & & \\ & t_2-t_1 & & \\ & & \ddots & \\ & & & t_n-t_{n-1} \end{pmatrix}$$

Thus Δ is Gaussian with uncorrelated (and thus independent), components with $N(0, t_j-t_{j-1})$ distrib. This proves that increments are indep, stationary (and Gaussian). $\square$

INVARIANCE PROPERTIES

Various transformations of a BM are still BM.
For example:

- Reflection of BM. If $(B_t)_{t\geq 0}$ is a BM then $(X_t)_{t\geq 0}$ where $X_t := -B_t$ is still a BM.

Indeed

0) $X_0 = -B_0 = 0$ a.s.

i) Take $t_0 < t_1 < \cdots < t_j < \cdots t_n$, we have $X_{t_j} - X_{t_{j-1}} = -B_{t_j} + B_{t_{j-1}} = -(B_{t_j} - B_{t_{j-1}})$ and thus indep. follows by indep of $B_{t_j} - B_{t_{j-1}}$

2) For $s \le t$, $X_{t+h} - X_{s+h} = -(B_{t+h} - B_{s+h})$
$$\sim -(B_t - B_s)$$
$$= X_t - X_s$$
for any $h \ge 0$

3) $X_t - X_s = -(B_t - B_s) \sim -N(0, t-s)$
$$= N(0, t-s)$$

4) Continuity follows trivially.

- Renewal. Let B_t be a BM and fix $a > 0$ arbitrary. Then $(W_t)_{t \ge 0}$, $W_t := B(t+a) - B(a)$ is again a BM.

0) $W_0 = B(a) - B(a) = 0$

1) $W_{t_j} - W_{t_{j-1}} = B(t_j + a) - \cancel{B(a)} - B(t_{j-1} + a) \cancel{+ B(a)}$
$$= \underbrace{B(t_j + a) - B(t_{j-1} + a)}_{\text{indep. since } B \text{ is BM}}$$

2) $W_{t+h} - W_{s+h} = B^{a+}_{t+h} - B^{a+}_{s+h} \underset{\text{R.t}}{\sim} B_t - B^{a+}_{s}$
$$= W_t - W_s$$

3) $W_t - W_s = B_{a+t} - B_{a+s} \sim N(0, t-s)$

4) trivial.

A consequence of independence of increments is that BM has no memory, in the sense that it has the Markov property.

Lemma. Let $(B_t)_{t\geq 0}$ be a BM. Let $W_t := B_{t+a} - B_a$. Then $(B_t)_{0\leq t\leq a}$ and $(W_t)_{t\geq 0}$ are indep., i.e., the σ-algebras generated by them are indep.:

$$\sigma(B_t : 0\leq t\leq a) =: \mathcal{F}_a^B \perp\!\!\!\perp \mathcal{F}_\infty^W := \sigma(W_t : 0\leq t < +\infty)$$

In particular $B_t - B_s \perp\!\!\!\perp \mathcal{F}_s^B$ for all $s\leq t$.

Proof. First note that, for arbitrary r.v.'s $X_0\, X_1 \dots X_n$, it is true that

(*) $$\sigma(X_j : j=0\dots n) = \sigma(X_0, X_j - X_{j-1} : j=1\dots n).$$

To see this note that X_0 and $X_j - X_{j-1}$, $j=1\dots n$, are measurable wrt $\sigma(X_j : j=0\dots n)$, this proves the inclusion $\supset$. The converse inclusion instead follows since $X_k = \left[\sum_{j=1}^{k} (X_j - X_{j-1})\right] + X_0$, $k=1 \dots n$, and thus X_k, $k=1\dots n$, are $\sigma(X_0, X_j - X_{j-1} : j=1\dots n)$ measurable. Hence the inclusion $\subset$ follows.

Take now

$$0 = r_0 < r_1 < \dots < r_m \le a = t_0 < t_1 < \dots < t_n.$$

We prove now that the class of events

$$\bigcup_{m\in\mathbb{N}} \bigcup_{r_0<r_1<\dots<r_m\le a} \sigma(B_{r_j} : j=1\dots m) \perp\!\!\!\perp \bigcup_{n\in\mathbb{N}} \bigcup_{0<t_1-a<t_2-a<\dots<t_n-a} \sigma(W_{t_j} : j=1\dots n) \quad (*)$$

Needed for the class being a π-system

are independent. Since they are π-systems it follows that the generated σ-Algebra are indep., i.e., $\mathcal{F}^B_a$ and $\mathcal{F}^W_\infty$ are indep..

The fact that (*) is true follows since

$$B_{r_1}-B_{r_0} \;\dots\; B_{r_m}-B_{r_{m-1}} \quad B_{t_1}-B_{t_0} \;\dots\; B_{t_n}-B_{t_{n-1}}$$

are indep. by indep of increments and we can use (*) to get

$$\sigma(B_{r_j}, j=1\dots m) \perp\!\!\!\perp \sigma(\overbrace{W_{t_k-t_0}}^{B_{t_n}-B_a}, k=1\dots n).$$

Since r_j, $j=1\dots m$, t_j, $j=1\dots n$ and n and m are all arbitrary, the result follows.

Finally, taking $a=r$, we see that

$$W(t-r) = B(t)-B(r)$$

which $\mathcal{F}^W_\infty$-measurable. Hence $B(t)-B(r) \perp \mathcal{F}^B_r$.

Another useful property of BM is time inversion.

— Time inversion. Let $(B_t)_{t\geq 0}$ be a BM and fix some time $a>0$. Then $W_t := B_{a-t} - B_a$, $t \in [0,a]$ is again a BM. This follows very much as the renewal property. Indeed

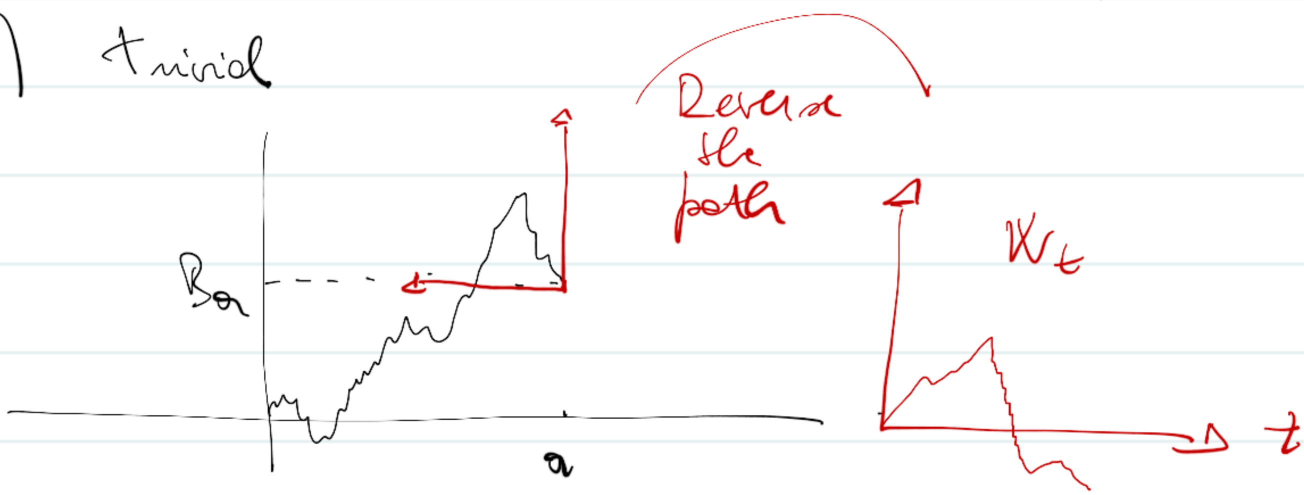
0) $W_0 = B_a - B_a = 0$

1) $W_{t_j} - W_{t_{j-1}} = B_{a-t_j} - B_a - B_{a-t_{j-1}} + B_a$
$$= -\underbrace{\left(B_{a-t_{j-1}} - B_{a-t_j}\right)}_{\text{indep. as } j=1\cdots n}$$

2) $W_{t+h} - W_{s+h} = -\left(B_{a-s-h} - B_{a-t-h}\right) \sim \left(B_{a-s} - B_{a-t}\right)$

3) $W_t - W_s = -\left(B_{a-s} - B_{a-t}\right) \sim -N(0, t-s)$
$$= N(0, t-s)$$

4) trivial



Here is instead the well known scaling property of BM.

Scaling. For all $c>0$ and $t\geq 0$, we have that

$$B_{ct} \sim \sqrt{c}\, B_t$$

In particular $\left(c^{-1/2} B_{ct}\right)_{t\geq 0}$ is again a BM.

↳ A different scaling does not change indep. of increments or continuity. Further $N(0, ct) = \sqrt{c}\, N(0, t)$

— BM in $\mathbb{R}^d$

The goal now is to make clear that a process in $\mathbb{R}^d$ $B_t = (B_t^1 \cdots B_t^d)$ is a BM iff all of the coordinate processes B_t^i, $i=1 \dots d$ are indep. 1-dim. BMs. Independence of processes is meant in the sense that two processes $(X_t)_{t\geq 0}$ and $(Y_t)_{t\geq 0}$ are indep. if

$$\mathcal{F}_\infty^X \perp \mathcal{F}_\infty^Y$$

where

$$\mathcal{F}_\infty^X := \sigma\left(\bigcup_{m\geq 1} \bigcup_{\substack{0\leq t_1 < \cdots \\ < t_m < +\infty}} \sigma(X_{t_1}, \dots X_{t_m}) \right)$$

Note that, as we already explained before, the σ-algebra $\mathcal{F}_s^X$ is generated by a π-system. Hence indep. of the σ-Algebras follows by indep of the π-systems. Hence we should prove

$$(X_{s_1}, \dots X_{s_n}) \perp (Y_{t_1}, \dots Y_{t_m})$$

for every $m, n \geq 1$, $s_1 < \dots < s_n$ and $t_1 < \dots < t_m$.

Indeed suppose that we prove

$(X_{s_1}, X_{s_2}) \perp (Y_{s_1}, Y_{s_2})$ for all $s_1, s_2 > 0$. $\circledast$

It is true that

$$X_s \perp Y_t \quad \text{for } s,t > 0 \text{ arbitrarily ?}$$

$$P(X_s \in B_1, Y_t \in B_2) =$$

$$= P(X_s \in B_1, X_t \in \mathbb{R}, Y_s \in \mathbb{R}, Y_t \in B_2)$$

by $\circledast$ $\quad = P(X_s \in B_1, X_t \in \mathbb{R}) \; P(Y_s \in \mathbb{R}, Y_t \in B_2)$

$$= P(X_s \in B_1) \quad P(Y_t \in B_2)$$

By generalizing this idea the result follows.

It will be usefull to use the following characterization of BM.

Lemma. Let $(X_t)_{t \geq 0}$ be a d-dim stochastic process. Then it satisfies 0) -- 3) iff, for all $n \geq 0$, and

$E e^{i\langle \xi, \underline{x}\rangle}$ $\underline{x} = \begin{bmatrix} X_{t_1} - X_{t_0} \\ \vdots \\ X_{t_m} - X_{t_{m-1}} \end{bmatrix}$

$0 = t_0 < t_1 \cdots < t_m$, and $\xi_0 \cdots \xi_m \in \mathbb{R}^d$,

$$E\left[\exp\left(i\sum_{j=1}^{m} \langle \xi_j, X_{t_j} - X_{t_{j-1}}\rangle + i\langle \xi_0, X_{t_0}\rangle\right)\right] \qquad (*)$$
$$= \exp\left(-\frac{1}{2}\sum_{j=1}^{m} |\xi_j|^2 (t_j - t_{j-1})\right).$$

Hence if X has continuous sample paths then it is a BM^d.

Proof Assume that X satisfies $0) \cdots 3)$. Since the characteristic function of a Gaussian r.v. with variance matrix $(t-s)\mathbb{1}_d$ and 0 expectation, is $\exp\left(-\frac{1}{2}(t-s)|\xi|^2\right)$, we get

$$E\left[\exp\left(i\sum_{j=1}^{m} \langle \xi_j, X_{t_j} - X_{t_{j-1}}\rangle + i\langle \xi_0, X_{t_0}\rangle\right)\right.$$
$$\overset{0)-1)}{=} \left(\prod_{j=1}^{m} E e^{i\langle \xi_j, X_{t_j} - X_{t_{j-1}}\rangle}\right) \underbrace{E e^{i\langle \xi_0, X_{t_0}\rangle}}_{=1 \text{ by } 0)}$$

$$\overset{2)}{=} \prod_{j=1}^{m} E\left[\exp\left(i\langle \xi_j, X_{t_j - t_{j-1}}\rangle\right)\right]$$

$$\overset{3)}{=} \prod_{j=1}^{m} \exp\left(-\frac{1}{2}(t_j - t_{j-1})|\xi_j|^2\right)$$

Conversely, assume that the statement (*) holds. Fix $k \in \{0 \dots n\}$ and pick in (*) $\xi_j = 0$ for every $j \neq k$. Then (*) reads

$$E e^{i\langle \xi_k, X_{t_k} - X_{t_{k-1}} \rangle} = e^{-\frac{1}{2}|\xi_k|^2 (t_k - t_{k-1})}$$

from which 3) follows.

But also

$$E e^{i\langle \xi_k, X_{t_k - t_{k-1}} \rangle} = e^{-\frac{1}{2}|\xi_k|^2 (t_k - t_{k-1})}$$

Hence

$$E e^{i\langle \xi_k, X_{t_k} - X_{t_{k-1}} \rangle} = E e^{i\langle \xi_k, X_{t_k - t_{k-1}} \rangle}$$

from which 2) follows.

We also have that 1) is verified by (*). Then by letting $n \to \infty$ 0) follows.

→ since (*) is the factoriz. of the ch functions of $X_{t_k} - X_{t_{k-1}}$ (Kac theorem)

It follows from previous result that if B_t is BM^d then the coordinate processes are 1-dim indep. BM's.

Corollary. Let B be a BM^d. Then the coordinate processes B^j, $j=1\dots d$, are indep. BM^1. $\square$

Proof. Since B is BM^d then it satisfies $\#$.
Use $\#$ for $\underbrace{\xi_j}_{\in\mathbb{R}^d} = \underbrace{z_j}_{\in\mathbb{R}} \underbrace{e_k}_{\in\mathbb{R}^d}$ where $e_k = \underbrace{[0 \cdots \overset{k}{1} \cdots 0]}_{d}^T$

for all $j = 1 \dots n$ and any arbitrary fixed $n \in \mathbb{N}$.
Then B^k satisfies $\#$ for all $z_1 \dots z_n \in \mathbb{R}$.
Continuity of paths of B^j follows from continuity of paths of B. Hence B^j is a BM^1 for every $j = 1 \dots d$. Now we should prove that these 1-dim BMs are indep. Hence we should verify that $(\mathcal{F}_t^{B^1})_{t\geq 0} \dots (\mathcal{F}_t^{B^d})_{t\geq 0}$ are indep., i.e., that

$$(B_{t_1}^k \dots B_{t_n}^k) \quad k=1,2\dots d \quad \text{are indep.}$$

for any choice of $n \in \mathbb{N}$ and $t_0 = 0 \leq t_1 \dots \leq t_n$.
Since each $B_{t_\ell}^k$, $\ell = 1 \dots n$, can be written as $B_{t_\ell}^k = \sum_{j=1}^{\ell} (B_{t_j}^k - B_{t_{j-1}}^k)$ it is enough to show that all the increments $(B_{t_j}^k - B_{t_{j-1}}^k)$, $j=1\dots n$, $k=1\dots d$ are indep. This follows again from $\#$ since

Let $\underbrace{\xi^k}_{\mathbb{R}^M} = (\underbrace{\xi_1^k}_{\in\mathbb{R}} \dots \underbrace{\xi_m^k}_{\in\mathbb{R}})$

$E e^{i\sum_1^d \langle \xi^k, \Delta\rangle}$ $\quad y^n = (B^k_{t_1} - B^k_{t_0}, \dots, B^k_{t_m} - B^k_{t_{m-1}})$

$$\mathbb{E} \exp\left(i \sum_{j=1}^m \sum_{k=1}^d \xi_j^k (B^k_{t_j} - B^k_{t_{j-1}})\right) = \mathbb{E} \exp\left(i \sum_{j=1}^m \langle \xi_j, B_{t_j} - B_{t_{j-1}} \rangle\right)$$

$$\overset{\#}{=} \exp\left(-\frac{1}{2} \sum_{j=1}^m |\xi_j|^2 (t_j - t_{j-1})\right)$$

$$= \exp\left(-\frac{1}{2} \sum_{j=1}^m \sum_{k=1}^d (\xi_j^k)^2 (t_j - t_{j-1})\right)$$

$$= \prod_{j=1}^m \prod_{k=1}^d \exp\left(-\frac{1}{2} (\xi_j^k)^2 (t_j - t_{j-1})\right)$$

$$= \prod_{j=1}^m \prod_{k=1}^d \mathbb{E} \exp\left(i \xi_j^k (B^k_{t_j} - B^k_{t_{j-1}})\right)$$

$$= \prod_{k=1}^d E e^{i\langle \xi^k, \Delta^k\rangle}$$

since the B_t^k are BM's. — since incr. for fixed k are indep. — for fixed k it is a BM

Independence thus follows.

The converse of this corollary is also true. Hence we can prove the following theorem.

Theorem. B is a BM^d iff the coordinate processes $B^1 \dots B^d$ are indep. BM^1.

Proof. By previous corollary we should prove only the if part. Hence we check that B is a BM^d

if B^j_t are BM^1 for $j=1 \dots d$.
then fix n, $t_0 = 0 < t_1 \dots < t_n < +\infty$, $\xi^k_j \in \mathbb{R}$, $j = 1 \dots n$, $k = 1 \dots d$. By assumption, each B^j satisfies ($\star$), i.e., for all $k = 1 \dots d$

$$E\left[\exp\left(i \sum_{j=1}^{n} \xi^k_j \left(B^k_{t_j} - B^k_{t_{j-1}}\right)\right)\right]$$

$$= \exp\left(-\frac{1}{2} \sum_{j=1}^{n} (\xi^k_j)^2 (t_j - t_{j-1})\right).$$

Multiplying both sides of the equalities above for $k = 1 \dots d$ and using indep. of B^j, $j = 1 \dots k$ we get

$$E\left[\prod_{k=1}^{d} \exp\left(i \sum_{j=1}^{n} \xi^k_j \left(B^k_{t_j} - B^k_{t_{j-1}}\right)\right)\right]$$

$$= \exp\left(-\frac{1}{2} \sum_{k=1}^{d} \sum_{j=1}^{n} (\xi^k_j)^2 (t_j - t_{j-1})\right)$$

which is exactly ($\star$) for $B = (B^1 \dots B^d)$ and $\xi_j = (\xi^1_j \dots \xi^d_j) \in \mathbb{R}^d$. The result follows. $\lrcorner$